

Notes on lecture 3

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1 Derivative of a vector function

It will be easy to understand this topic if we interpret the vector function in terms of space curves.

Definition 1.0.1. Suppose $\vec{r}(t)$ is a vector function, where $t \in [a, b]$. Then the **derivative** of $\vec{r}(t)$, denoted by $\frac{d\vec{r}}{dt}$ is given by:

$$\frac{d\vec{r}}{dt} = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \quad (+)$$

provided the limit exists. If the limit exists, then we say that the derivative of the vector function $\vec{r}$ exists and it is given by (+). $\square$

1.1 Geometric interpretation of the derivative

The secant approach: Suppose we are considering a space curve. Suppose at time t , the particle is at the point P on the space curve. And suppose at time $t+h$, the particle is at the point Q on the space curve. Then the vector $\overrightarrow{PQ}$ equals $\vec{r}(t+h) - \vec{r}(t)$. See part (a) of Figure 1.1.1 for an illustration. If we divide the vector $\vec{r}(t+h) - \vec{r}(t)$ by h (where h is a very small real number, that is, $\frac{1}{h}$ is a very big real number), then in fact, we are multiplying the vector $\overrightarrow{PQ}$ by a scalar $\lambda (= \frac{1}{h})$, which is a very big real number. Then definitely, the new vector $\frac{\vec{r}(t+h) - \vec{r}(t)}{h}$ is going to have the same direction as $\overrightarrow{PQ}$. But its length will be much much bigger than the length of $\overrightarrow{PQ}$. This vector $\frac{\vec{r}(t+h) - \vec{r}(t)}{h}$ passing through P and Q is called the **secant vector**. See Figure 1.1.1 for an illustration.

As we let $h \rightarrow 0$, the secant vector will approach the tangent to the curve at the point P (See part (b) of Figure 1.1.1). This finishes the geometrical interpretation.

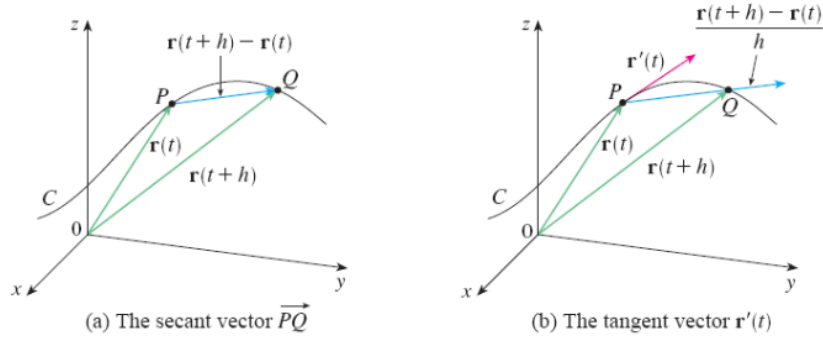


Figure 1.1.1: Derivative of a vector function

1.2 Notation

$\frac{d\vec{r}}{dt}$ is also denoted by $\vec{r}'(t)$. If $\vec{r}'(t) \neq \vec{0}$, then the derivative defines a tangent to the curve C at the point P .

We often denote $\vec{r}'(t)$ by $\vec{T}$. The **unit tangent vector** along the direction of the tangent is denoted by $\hat{t}$, where

$$\hat{t} = \frac{\vec{T}}{|\vec{T}|} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Recall that if

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

is a vector function, then its derivative $\vec{r}'(t)$ is given by

$$\vec{r}'(t) = \frac{d\vec{r}}{dt} = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}. \quad (*)$$

Theorem 1.2.1. If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ is a vector function, then

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle,$$

provided each of the component functions of $\vec{r}(t)$ is a differentiable function.

PROOF: It follows from (*) above that

$$\vec{r}'(t) = \langle \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}, \lim_{h \rightarrow 0} \frac{g(t+h) - g(t)}{h}, \lim_{h \rightarrow 0} \frac{h(t+h) - h(t)}{h} \rangle.$$

But now, it follows from single variable calculus that the right hand side of the above equation is equal to $\langle f'(t), g'(t), h'(t) \rangle$, provided the functions f, g, h are differentiable at t . $\square$

Remark 1.2.2. Whenever we need to find out the derivative of a vector function in terms of its components, we simply need to find out the derivatives of its components and place them as the components of the derivative vector function, provided the component functions are differentiable.

Example 1.2.3. If $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$, then $\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$. $\square$

Some important properties of the derivative are listed in Theorem 1.2.4 below.

Theorem 1.2.4. Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function. Then

1. $\frac{d}{dt} [\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
2. $\frac{d}{dt} [c\mathbf{u}(t)] = c\mathbf{u}'(t)$
3. $\frac{d}{dt} [f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
4. $\frac{d}{dt} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
5. $\frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
6. $\frac{d}{dt} [\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$ (Chain Rule)

Proof of (5) in Theorem 1.2.4: By definition, we have

$$\frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \lim_{h \rightarrow 0} \left[\frac{\mathbf{u}(t+h) \times \mathbf{v}(t+h) - \mathbf{u}(t) \times \mathbf{v}(t)}{h} \right].$$

It is easy to see that the right hand side can be rewritten as

$$\lim_{h \rightarrow 0} \left[\frac{\mathbf{u}(t+h) \times \mathbf{v}(t+h) - \mathbf{u}(t+h) \times \mathbf{v}(t) + \mathbf{u}(t+h) \times \mathbf{v}(t) - \mathbf{u}(t) \times \mathbf{v}(t)}{h} \right].$$

If we now bring the limit inside, then from the properties of cross product (studied in earlier lectures), we can say that the above expression equals

$$\lim_{h \rightarrow 0} \mathbf{u}(t+h) \times \lim_{h \rightarrow 0} \frac{\mathbf{v}(t+h) - \mathbf{v}(t)}{h} + \lim_{h \rightarrow 0} \frac{\mathbf{u}(t+h) - \mathbf{u}(t)}{h} \times \mathbf{v}(t),$$

which in turn equals $\mathbf{u}(t) \times \mathbf{v}'(t) + \mathbf{u}'(t) \times \mathbf{v}(t)$. $\square$

We will now use part (4) of Theorem 1.2.4 above to derive an important result, which is going to be very useful later on.

Result: If $\mathbf{u}$ is a vector of constant length, then $\mathbf{u}$ and $\mathbf{u}'$ are orthogonal to each other.

Proof of the result: Let $|\mathbf{u}| = a$, where a is a constant. Then $|\mathbf{u}|^2 = a^2$. This implies that $\mathbf{u} \cdot \mathbf{u} = a^2$. Now taking derivative, we get

$$\frac{d}{dt}[\mathbf{u} \cdot \mathbf{u}] = 0.$$

By part (4) of Theorem 1.2.4, we then have:

$$\mathbf{u} \cdot \mathbf{u}' + \mathbf{u}' \cdot \mathbf{u} = 0.$$

But we know that vector dot product is commutative. So the above equation implies

$$2\mathbf{u} \cdot \mathbf{u}' = 0 \Rightarrow \mathbf{u} \cdot \mathbf{u}' = 0.$$

This implies that $\mathbf{u}$ and $\mathbf{u}'$ are orthogonal to each other. $\square$

2 Vector integration

Recall from single variable calculus that to compute the integral $\int_a^b f(t)dt$ of a real valued function f over an interval $[a, b]$, we first divided the interval $[a, b]$ into sub intervals, then we considered the areas of the strips or rectangles formed by taking certain heights (dependent on the value of f) over those sub intervals, and then we added those areas. And as the number of sub intervals of the interval $[a, b]$ increases (or tend to ∞), the limit of the sum of the areas of all those strips was defined to be the integral $\int_a^b f(t)dt$.

We now have the following definition for the integral of a vector function:

Definition 2.0.1. Let $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ be a vector function. Then

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

□

Let us now state (without proof) the **fundamental theorem of integral calculus** for vector functions (as Theorem 2.0.2 below).

Theorem 2.0.2. Suppose $\vec{r}(t)$ is a continuous vector function such that $\vec{r}(t) = \vec{R}'(t)$ for some other vector function $\vec{R}(t)$. [In such a case, $\vec{R}(t)$ is called an **antiderivative** of $\vec{r}(t)$.] Then

$$\int_a^b \vec{r}(t) dt = \vec{R}(b) - \vec{R}(a).$$

3 Arc length and curvature

Let us first have a quick glimpse at how we are going to find arc length in cartesian coordinates. Suppose we consider a curve C in a 2 dimensional plane on an interval $[a, b]$. Assume that C is the graph of a function of one variable $y = y(x)$ on the interval $[a, b]$. Let $x_0 = a$ and $x_n = b$. Let us divide the interval $[a, b]$ into some sub intervals of the form $[x_{i-1}, x_i]$ ($1 \leq i \leq n$) by introducing some points $x_1, \dots, x_{n-1}$ in between the points a and b . Note that these points in between need not have to be equidistant from each other. Consider the interval $[x_{i-1}, x_i]$ for any $i \in \{1, \dots, n\}$. For $i \in \{1, \dots, n\}$, let P_i denote the point $(x_i, y(x_i))$ in the xy -plane. Join the points P_{i-1} and P_i for each i by a straight line. Then we can easily see that the length L of the curve C can be approximated by the sum $\sum_{i=1}^n |\overline{P_{i-1}P_i}|$. That is,

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |\overline{P_{i-1}P_i}|.$$

See Figure 3.0.1 for an illustration.

Let $\Delta x_i = x_i - x_{i-1}$ and $\Delta y_i = y(x_i) - y(x_{i-1})$. Observe that $|\overline{P_{i-1}P_i}| = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$. Then we have from the above equation that

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}.$$

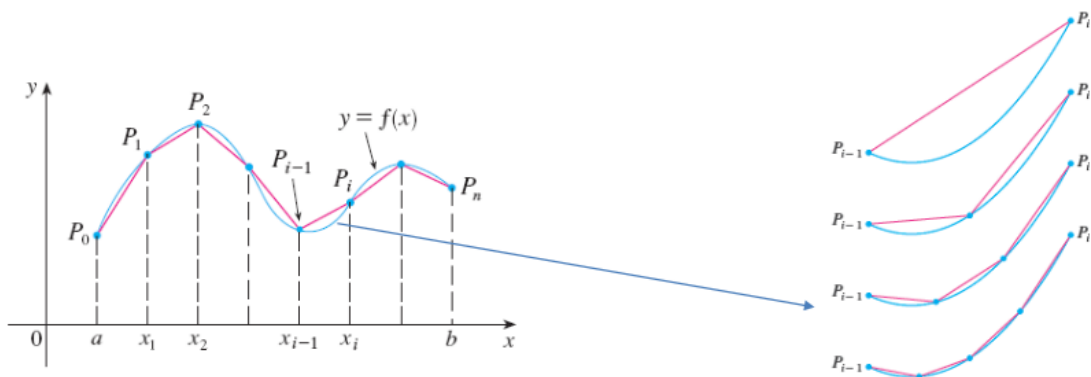


Figure 3.0.1: The arc length

Now if y is a differentiable function of the single variable x , then by mean value theorem (learnt in single variable calculus), we get

$$\Delta y_i = y(x_i) - y(x_{i-1}) = (x_i - x_{i-1})y'(x_i^*) = \Delta x_i y'(x_i^*),$$

for some $x_i^* \in (x_{i-1}, x_i)$. Then we can write

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + \{y'(x_i^*)\}^2} \Delta x_i, \end{aligned}$$

which in turn equals $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$. That is, we have

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (**)$$

The formula given in equation (**) above is known as the **arc length formula**, provided y' is continuous on $[a, b]$.

3.1 Arc length when the curve is parametrized by the variable t

Consider the plane curve C in Figure 3.1.1. Compare this curve C in Figure 3.1.1 with the curve in Figure 3.0.1. Observe that the curve C in Figure 3.1.1 is NOT the graph of a function of one variable, but the curve in

Figure 3.0.1 is so. This is because, in Figure 3.1.1, the curve C is drawn in a twisted way. If we draw a vertical line at the point where the curve C bends (or takes twist), then that vertical line is going to cut the curve C at more than one points. The curve C in Figure 3.1.1 is not the graph of a function of one variable. But still, there is a way out for finding the length of such a curve. And this is done by **parametrizing the curve C** by a new variable t .

Assume that the curve C is drawn in the xy -plane in such a way that the

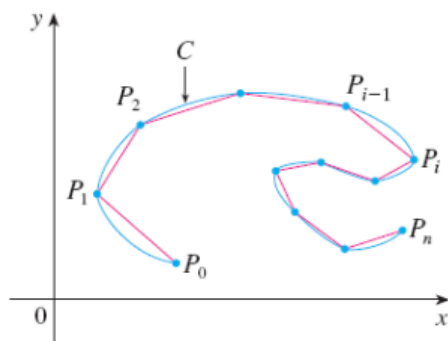


Figure 3.1.1: Arc length when the curve is parametrized by t

initial point P_0 of the curve has x -coordinate a and the final point P_n has x -coordinate b . Suppose we consider any point P on the curve C whose coordinates are (x, y) and we join the point P with the origin O . Say, $\vec{r} = \overrightarrow{OP}$. We are going to define $\vec{r}$ as $\langle x(t), y(t) \rangle$ (in terms of the variable t). That is,

$$\vec{r}(t) = \langle x(t), y(t) \rangle .$$

Here, we are considering an interval $[\alpha, \beta]$ (where the variable t lies). Let us assume that x and y are continuous functions of the single variable t .

Let us now divide the interval $[\alpha, \beta]$ (in which the parameter t lies) into some sub intervals: Let $t_0 = \alpha$, $t_n = \beta$, and $t_1, \dots, t_{n-1}$ be some intermediate points (which are not necessarily equidistant) on the interval $[\alpha, \beta]$. Through the points $t_0, t_1, \dots, t_n$, we will get hold of some points on the curve C in the following way: For each $i \in \{0, \dots, n\}$, let $P_i = (x(t_i), y(t_i))$ denote the corresponding point on the curve C . So we will get the points $P_0, P_1, \dots, P_n$ on the curve C . And corresponding to the points P_{i-1} and P_i ,

we have the interval $[t_{i-1}, t_i] \subseteq [\alpha, \beta]$. Let us join each pair of consecutive points (P_{i-1}, P_i) , where $i \in \{1, \dots, n\}$ by straight lines (that is, P_0 and P_1 , P_1 and P_2 , and so on). Then (as before), we will have

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i| = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2},$$

where L denotes the length of C . Now,

$$\Delta x_i = x_i - x_{i-1} = x(t_i) - x(t_{i-1})$$

Applying the mean value theorem, we get

$$x(t_i) - x(t_{i-1}) = (t_i - t_{i-1})x'(t_i^*) = \Delta t_i x'(t_i^*),$$

where $t_i^* \in (t_{i-1}, t_i)$ and $\Delta t_i = t_i - t_{i-1}$. Similarly, we get

$$\Delta y_i = \Delta t_i y'(t_i^{**}),$$

where $t_i^{**} \in (t_{i-1}, t_i)$. It hence follows that

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{(x'(t_i^*))^2 + (y'(t_i^{**}))^2} \Delta t_i \\ &= \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt. \end{aligned}$$

Here x' and y' should be continuous on $[\alpha, \beta]$ and the curve C should be traversed exactly once as t increases from α to β .

Remark 3.1.1. Whenever we consider a space curve or a plane curve in terms of a vector function, we should always consider the interval of the parameter t in such a way that it is in the increasing direction, so that we are able to track the curve.

In this way, we can find the length of a curve C in a plane, where the curve C has been parametrized by the variable t .

3.2 Arc length of a curve in space

Suppose now that we have a curve in the 3 dimensional space. Say, an arbitrary point P on the curve has coordinates $(x(t), y(t), z(t))$. So x, y, z are 3 functions from an interval $[\alpha, \beta]$ to the real line $\mathbb{R}$. The position vectors of the initial and the end points of the curve are then $\vec{r}(\alpha)$ and $\vec{r}(\beta)$ respectively. Then applying the same methodology (as in the 2 dimensional plane curve case), we get

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt. \quad (+)$$

Theorem 3.2.1. $L = \int_{\alpha}^{\beta} |\vec{r}'(t)| dt$, L denotes the length of a curve in space which is parametrized by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$ where $\alpha \leq t \leq \beta$.

PROOF: Observe that $\vec{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$. It is then clear that the quantity $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$ in equation (+) is nothing but the magnitude $|\vec{r}'(t)|$ of $\vec{r}'(t)$. So, we have (from (+)):

$$L = \int_{\alpha}^{\beta} |\vec{r}'(t)| dt.$$

□

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